

Grade 6 Accelerated
Unit 13
Lesson 12 Homework

Name Answer Key
Date _____
Period _____

Write an equivalent expression using the associative, commutative, or distributive property. State the property used. **Answers may vary**

1. $7x - \frac{3}{7}(14x - 21)$

$7x - 6x + 9$

distributive property

2. $(16x + 13) - 2x + 5$

$16x - 2x + 13 + 5$

commutative property

3. $\left(-\frac{2}{3}y + 8\right) \cdot 9$

$\left(8 - \frac{2}{3}y\right) \cdot 9$

commutative property

4. $-8\left(\frac{3}{4}x + 2\right) + (4 - 2x)$

$-6x - 16 + (4 - 2x)$

distributive property

5. If expressions are equivalent, then they will

- ☒ always
- ☐ sometimes
- ☐ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.7x - 0.1(4x - 2)$ and $7x - (4x - 2)$

$$0.7x - 0.4x + 0.2 \text{ and } 3x + 2$$

$$0.3x + 0.2$$

$$0.3(2) + 0.2$$

$$0.6 + 0.2$$

$$= 0.8$$

$$3(2) + 2$$

$$6 + 2$$

$$= 8$$

not equivalent

7. $(5x + 12) - 6$ and $7x - 2(x - 3)$

$$5x + 6$$

$$5(2) + 6$$

$$16$$

$$7x - 2x + 6$$

$$5x + 6$$

$$5(2) + 6$$

$$16$$

equivalent

8. $2\frac{1}{5}x - \frac{3}{10} - x$ and $\frac{6}{5}\left(x - \frac{2}{5}\right)$

$$\frac{11}{5}x - \frac{3}{10} - \frac{5}{5}x$$

$$\frac{6}{5}x - \frac{3}{10}$$

$$\frac{6}{5} \cdot 5 - \frac{3}{10}$$

$$6 - \frac{3}{10} = 5\frac{7}{10}$$

$$\frac{6}{5}x - \frac{12}{25}$$

$$\frac{6}{5} \cdot 5 - \frac{12}{25}$$

$$6 - \frac{12}{25}$$

$$5\frac{13}{25}$$

not equivalent

9. $0.2 + 0.1(14x - 2)$ and $2x - \left(\frac{3}{5}x - \frac{1}{5}\right)$

$$0.2 + 1.4x - 0.2$$

$$1.4x$$

$$1.4(2) = 2.8$$

$$2x - \frac{3}{5}x + \frac{1}{5}$$

$$\frac{10}{5}x - \frac{3}{5}x + \frac{1}{5}$$

$$\frac{7}{5}x + \frac{1}{5}$$

$$\frac{7}{5} \cdot 2 + \frac{1}{5} \rightarrow \frac{14}{5} + \frac{1}{5} = 3$$

not equivalent

10. Expressions that are equal are also equivalent.

☒ True

☐ False